

Graph Data Structure

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Graph Data Structure is a collection of **nodes** connected by **edges**. It's used to represent relationships between different entities.

If you are looking for topic-wise list of problems on different topics like DFS, BFS, Topological Sort, Shortest Path, etc., please refer to [Graph Algorithms](#).

Basics of Graph:

- [Introduction to Graphs](#)
- [Graph and its representations](#)
- [Transpose graph](#)

Easy Problems

- [Breadth First Traversal for a Graph](#)
- [Depth First Traversal for a Graph](#)
- [Shortest path in an unweighted graph](#)
- [0-1 BFS \(Shortest Path in a Binary Weight Graph\)](#)
- [Count number of trees in a forest](#)
- [Topological Sorting](#)
- [All topological sorts of a Directed Acyclic Graph](#)
- [Kahn's Algorithm for Topological Sorting](#)
- [Sum of dependencies in a graph](#)
- [Maximum Weight Node](#)

Medium Problems

- [Transitive Closure of a Graph using DFS](#)
- [Rotten Tomatoes](#)
- [Islands in a Graph](#)
- [Flood Fill](#)
- [Check for Bipartite](#)
- [Word Ladder](#)
- [Snakes and Ladder](#)

- [Water Jug problem](#)
- [Shortest Path in Binary Matrix](#)
- [Clone a Graph](#)
- [Detect Cycle in a Directed Graph](#)
- [Detect cycle in an Undirected graph](#)
- [Detect a negative cycle in a Graph | \(Bellman Ford\)](#)
- [Cycles of length n in an undirected and connected graph](#)
- [Detecting negative cycle using Floyd Warshall](#)
- [Clone a Directed Acyclic Graph](#)
- [Introduction to Disjoint Set Data Structure or Union-Find Algorithm](#)
- [Union By Rank and Path Compression in Union-Find Algorithm](#)
- [Dijkstra's shortest path algorithm](#)
- [Bellman-Ford Algorithm](#)
- [Floyd Warshall Algorithm](#)
- [Johnson's algorithm for All-pairs shortest paths](#)
- [Multistage Graph \(Shortest Path\)](#)
- [Karp's minimum mean \(or average\) weight cycle algorithm](#)
- [Find minimum weight cycle in an undirected graph](#)
- [Prim's Minimum Spanning Tree \(MST\)](#)
- [Kruskal's Minimum Spanning Tree Algorithm](#)
- [Difference between Prim's and Kruskal's algorithm for MST](#)
- [Applications of Minimum Spanning Tree Problem](#)
- [Minimum cost to connect all cities](#)
- [Minimum Product Spanning Tree](#)
- [Reverse Delete Algorithm for Minimum Spanning Tree](#)
- [Boruvka's algorithm for Minimum Spanning Tree](#)
- [Maximum edges that can be added to DAG so that it remains DAG](#)
- [Longest Path in a Directed Acyclic Graph](#)
- [Topological Sort of a graph using departure time of vertex](#)
- [Find Itinerary from a given list of tickets](#)
- [Eulerian path and circuit](#)
- [Fleury's Algorithm for printing Eulerian Path or Circuit](#)
- [Strongly Connected Components](#)
- [Count all possible walks from a source to a destination with exactly k edges](#)

- [Euler Circuit in a Directed Graph](#)
- [Paths to travel each nodes using each edge \(Seven Bridges of Königsberg\)](#)
- [Length of shortest chain to reach the target word](#)
- [Dynamic Connectivity | Set 1 \(Incremental\)](#)
- [Max Flow Problem Introduction](#)
- [Ford-Fulkerson Algorithm for Maximum Flow Problem](#)
- [Find maximum number of edge disjoint paths between two vertices](#)
- [Maximum Bipartite Matching](#)
- [Channel Assignment Problem](#)
- [Karger's Algorithm- Set 1- Introduction and Implementation](#)
- [Dinic's algorithm for Maximum Flow](#)
- [Find length of the largest region in Boolean Matrix](#)
- [A Peterson Graph Problem](#)
- [Clone an Undirected Graph](#)
- [Graph Coloring \(Introduction and Applications\)](#)
- [Erdos Renyl Model \(for generating Random Graphs\)](#)
- [Chinese Postman or Route Inspection | Set 1 \(introduction\)](#)
- [Hierholzer's Algorithm for directed graph](#)
- [Check whether a given graph is Bipartite or not](#)
- [Snake and Ladder Problem](#)
- [Boggle \(Find all possible words in a board of characters\)](#)
- [Hopcroft Karp Algorithm for Maximum Matching-Introduction](#)
- [Minimum Time to rot all oranges](#)
- [Construct a graph from given degrees of all vertices](#)
- [Determine whether a universal sink exists in a directed graph](#)
- [Number of sink nodes in a graph](#)
- [Two Clique Problem \(Check if Graph can be divided in two Cliques\)](#)

Hard Problems

- [Pacific Atlantic Water Flow](#)
- [Total number of Spanning Trees in a Graph](#)
- [Longest Path in a Directed Acyclic Graph](#)
- [Bridges in a graph](#)
- [Articulation Points \(or Cut Vertices\) in a Graph](#)

- [Biconnected Components](#)
- [Find if an array of strings can be chained to form a circle](#)
- [Tarjan's Algorithm to find strongly connected Components](#)
- [Find minimum s-t cut in a flow network](#)
- [Traveling Salesman Problem \(TSP\) Implementation](#)
- [Vertex Cover Problem | Set 1 \(Introduction and Approximate Algorithm\)](#)
- [K Centers Problem | Set 1 \(Greedy Approximate Algorithm\)](#)

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